Sentimental Analysis Using Social Media Comments

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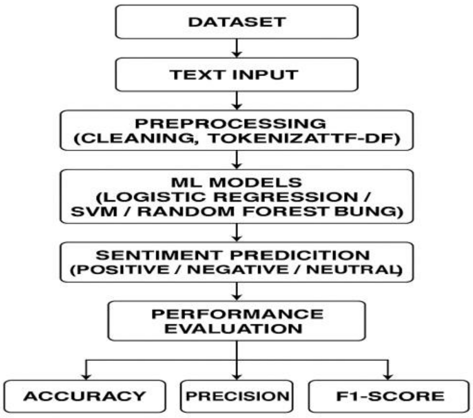
Abstract

This study uses advanced machine learning methods like Logistic Regression, Support Vector Machines (SVM), Random Forest, and Boosting models to classify the sentiment of social media comments into positive, negative, or neutral categories. It operates on Reddit comment data, focusing on cleaning, preprocessing, and extracting meaningful text features using techniques like TF-IDF. The models are developed to accurately capture the sentiment trends in user-generated content and understand emotional patterns across different topics.

Architecture Diagram



Significance of the Project

Prioritizing emotional well-being in SDGs.
Detecting public sentiment early.
Spreading awareness through social media.
Using ML to improve sentiment analysis.
Supporting research and innovation in NLP.
Helping organizations understand public opinion.
Promoting better decision-making through

Conclusion

The research provides an effective and interpretable method to classify user sentiment from social media comments using traditional machine learning models and text-based feature extraction. Our experiments show improved sentiment prediction accuracy and address common challenges such as class imbalance, noisy data, and model evaluation. By solving these issues, our model offers a scalable approach for understanding public emotions and supporting better decision-making based on social media insights.

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Conference/Journal Publication Details (Mandatory)

4th 2025 IEEE World Conference on Applied Intelligence and Computing : Submission (2095) has been created. Inbox ×

Microsoft CMT <noreply@msr-cmt.org>
to me

Sun 27 Apr, 23:58 (3 minutes ago)

Hello,

The following submission has been created.

Track Name: AIC2025

Paper ID: 2095

Paper Title: Sentimental analysis using social media comments

Abstract:
Social media has been a key source of public opinion, enabling researchers to tap into spontaneous and unmediated user-generated content. This paper investigates sentiment classi f ication from Reddit comments to determine user attitudes in pre defined categories. The methodology starts with data cleaning and preprocessing text data followed by feature extraction using Term Frequency-Inverse Document Frequency (TF-IDF). To handle class imbalance—a typical problem in real-world datasets—we employ oversampling using the RandomOverSampler algorithm. Preprocessed data is then used to train a Linear Support Vector Classifier (LinearSVC), which is selected based on its ability to cope with high-dimensional text features. Model assessment is done using accuracy, confusion matrices, and classification reports. Results show that the employment of TF-IDF combined with oversampling has a significant impact on enhancing model performance, particularly in skewed data environments. This paper presents a practical, scalable approach of performing sentiment analysis on social media data.

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Primary Subject Area: Computational intelligence

Secondary Subject Areas:
Intelligent Systems (Hardware & Software)